AARUSH PANDA

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EDUCATION

University of California, Berkeley | Expected May 2027 — **Berkeley, CA**
B.S. Mechanical Engineering (Aerospace Concentration), Entrepreneurship & Technology Certificate — GPA: 3.3
Activities: Undergraduate Research, VTOL @ Berkeley; Captain, Division I Club Frisbee

PROFESSIONAL EXPERIENCE

V2X — **Indianapolis, IN**
Mechanical Engineering Intern — *May 2026 – Jul 2026*
- Standardized server-rack housing for electrical components using COTS 80/20, enabling fabrication and program reuse
- Evaluated generator fit within equipment enclosure, resolving packaging and interface constraints for system integration
- Designed custom steel brackets and fuel-cell hardware to meet customer constraints, enabling on-time critical delivery
- Released production-ready PDU drawings in Creo compliant with MIL-STD-810 requirements enabling fabrication
- Supported missile launcher build and integration, enabling on-time customer demonstration and 3 potential contract leads

Naval Nuclear Laboratory - Bettis Atomic Power Laboratory (DOE L Clearance) — **Pittsburgh, PA**
Mechanical Engineering Intern — *Jun 2025 – Aug 2025*
- Designed and tolerance-stacked a payload-leveling mechanism for spent-fuel exams, achieving ±0.1° levelling accuracy
- Evaluated structural integrity via hand calculations and Math-CAD analyses, verifying safety factors against design loads
- Engineered ASME Y14.5-compliant Creo assemblies, improving assembly repeatability across product lifecycle
- Implemented DFM changes to reduce feature complexity and tolerance variation, cutting prototype lead time by 25%

VTOL (Vertical Takeoff and Landing) @ Berkeley — **Berkeley, CA**
President and Lead Mechanical Engineer — *Aug 2023 – Dec 2025*
- Led end-to-end mechanical design of Berkeley's 1st human-carrying eVTOL aircraft, delivering flight-critical hardware
- Designed, integrated, and tested carbon-fiber landing gear and tiltrotor mechanisms in SolidWorks and Fusion 360
- Calculated FAA-compliant load cases and verified structural integrity with safety factors under worst-case loads
- Optimized carbon-fiber structures through FEA and prototyping, reducing mass by 15% and improving thrust-to-weight

Berkeley Risk and Security Lab — **Berkeley, CA**
Undergraduate Researcher — *Sep 2024 – May 2026*
- Performed stress and fracture analyses on pressure vessels across failure modes, guiding non-proliferation material choices
- Mapped structural load paths across reactor assemblies to identify load-bearing components constraining physical access
- Analyzed failure modes of multi-sensor counter-UAS systems supporting reliable operation under contested conditions

PROJECTS

Custom Tilt-Rotor UAV (Link)
- Led end-to-end mechanical and aerodynamic design, build, and test of a custom tiltrotor UAV
- Performed ANSYS CFD across hover and cruise conditions, reducing wake interference and drag
- Designed and fabricated carbon-fiber primary structures and tilt mechanisms, validating strength under operational loads
- Tuned ArduPilot flight controls and validated performance via flight testing, achieving stable hover-to-cruise transitions

Robotic Shoe Shelver System (Link)
- Designed automated load-cell triggered actuator system enabling hands-free shoe removal
- Integrated load cell, actuator driver, and sensor feedback to prevent overload and repeated actuator-induced failures
- Built event-driven MicroPython and ESP-NOW logic enforcing safe sequencing between electro-mechanical subsystems

HushLight (Link)
- Designed MEMS microphone control and LED circuitry, validating threshold-based noise detection and visual feedback
- Integrated electronics and wiring into compact 3D-printed housing, resolving packaging and assembly constraints
- Assembled and bench-tested prototype, validating repeatable activation, robustness, and end-to-end sensing performance

CrutchGuard (Link)
- Identified crutch injury mechanisms affecting 12% of users and developed an integrated mitigation mechanism
- Designed and built functional prototype, applying GD&T to achieve ±0.01 mm dimensional accuracy
- Iterated fit, geometry, and material selection through physical prototypes, improving assembly and user interface

SKILLS & INTERESTS

Hardware: Structural Analysis, FEA, GD&T, DFM, DFMEA, Additive Manufacturing (FDM/SLA), Machining, Composites
Software: SolidWorks, Creo, Siemens NX, OnShape, Fusion 360, ANSYS, MATLAB, Python, Embedded Systems
Interests: Ultimate Frisbee, Soccer, Defense Strategy, Weightlifting, Entrepreneurship